

Memory

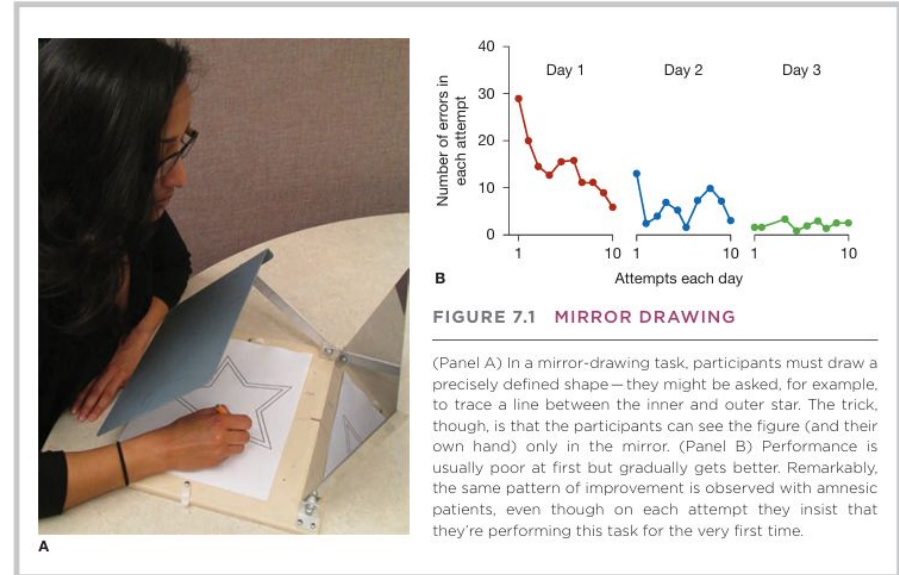
Introduction to Brain and Cognition
Spring 2026

Revisiting Amnesia

Clive Wearing

1. Linguistics intact
2. Piano playing intact
3. Remembers wife
4. Autobiographical memory deficit
5. Incapable of forming new lasting memories
6. Working memory short window

There clearly are different memory types



Hierarchy of Memory Types

Language

- Region involved: Neocortex, Broca/Wernicke network, left temporal lobe
- Type: Explicit/Declarative - Semantic memory

Piano

- Region involved: Basal Ganglia, Cerebellum, motor cortex
- Type: Implicit/Nondeclarative - Procedural

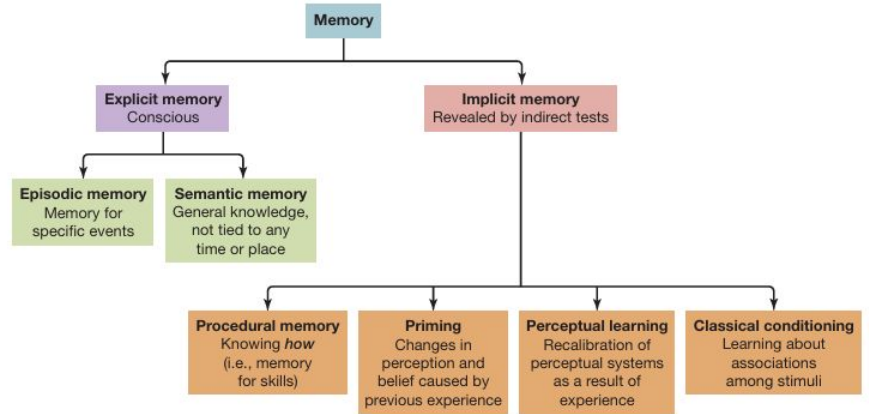
Wife

- Region Involved: **Hippocampus**, amygdala, Neocortex (old autobiographical traces)
- Type: Explicit - Episodic + Semantic + Emotional

Another female patient (Schacter, 1996)

1. **Suffered damage (from encephalitis) to front of temporal lobe**
2. **Affected - Common words, important historical events, fundamental traits of animate & inanimate objects (Semantic)**
3. **Remember - Wedding, honeymoon, father's illness & death, other specific episodes (Episodic)**

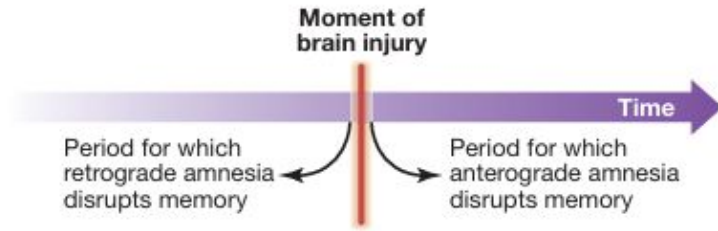
FIGURE 7.11 HIERARCHY OF MEMORY TYPES



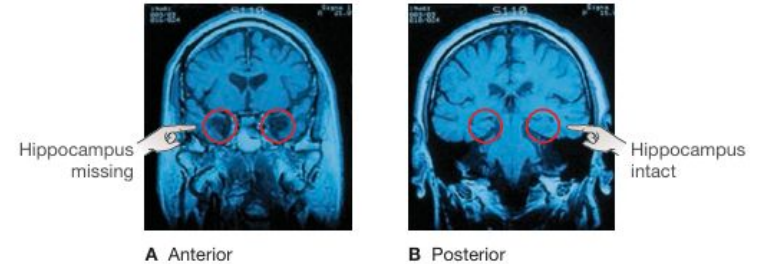
In our discussion, we've distinguished two types of memory — explicit and implicit. However, there are reasons to believe that each of these categories must be subdivided further, as shown here. Evidence for these subdivisions includes functional evidence (the various types of memory follow different rules) and biological evidence (the types depend on different aspects of brain functioning).

Types of Amnesia

FIGURE 7.12 RETROGRADE AND ANTEROGRADE AMNESIA



Retrograde amnesia disrupts memory for experiences *before* the injury, accident, or disease that triggered the amnesia. Anterograde amnesia disrupts memory for experiences *after* the injury or disease. Some patients suffer from both retrograde and anterograde amnesia.



H.M.'S BRAIN

For many years, researchers thought that surgery had left H.M. with no hippocampus at all. These MRI scans of his brain show that the surgery did destroy the anterior portion of the hippocampus (the portion closer to the front of the head; Panel A) but not the posterior portion (closer to the rear of the head; Panel B).

HM - remembered all from before surgery (HC removed) - ?

Clive Wearing - ? Korsakoff's - ?

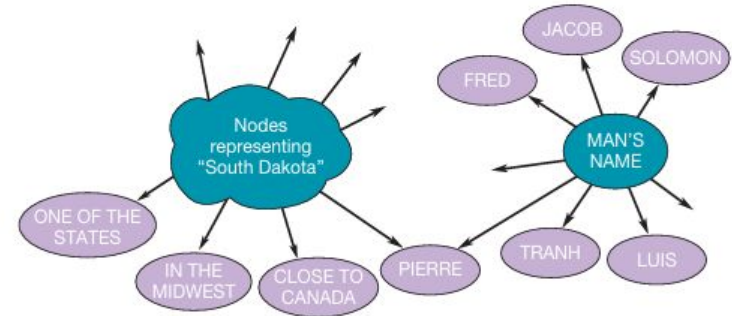
Memory Connections

- South Dakota - Pierre
- Man's name - Pierre

Multiple memory connections - retrieval paths - strengthen's retrieval

Repeated access of a certain memory, strengthens the connections of that memory trace & later retrieval

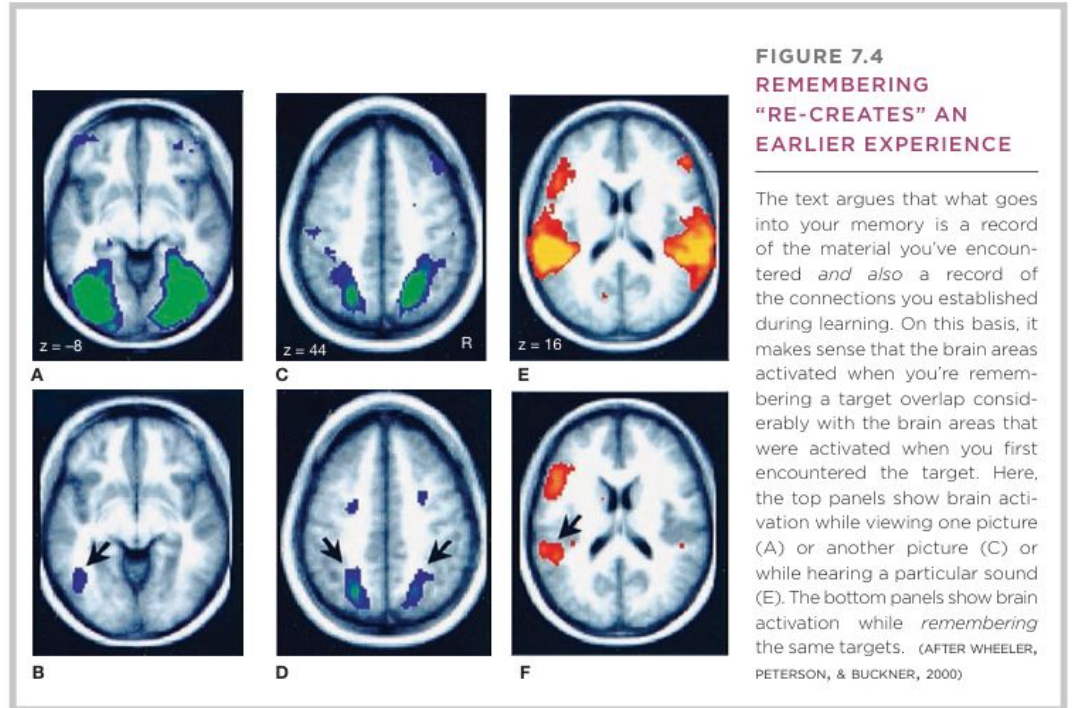
FIGURE 7.5 ACTIVATION OF A NODE FROM TWO SOURCES



A participant is asked, "What is the capital of South Dakota?" This activates the SOUTH DAKOTA nodes, and activation spreads from there to all of the associated nodes. However, it's possible that the connection between SOUTH DAKOTA and PIERRE is weak, so PIERRE may not receive enough activation to reach threshold. Things will go differently, though, if the participant is also given the hint "South Dakota's capital is also a man's name." Now, the PIERRE node will receive activation from two sources: the SOUTH DAKOTA nodes and the MAN'S NAME node. With this double input, it's more likely that the PIERRE node will reach threshold. This is why the hint ("man's name") makes the memory search easier.

Context Reinstatement

Remember encoding
specificity?



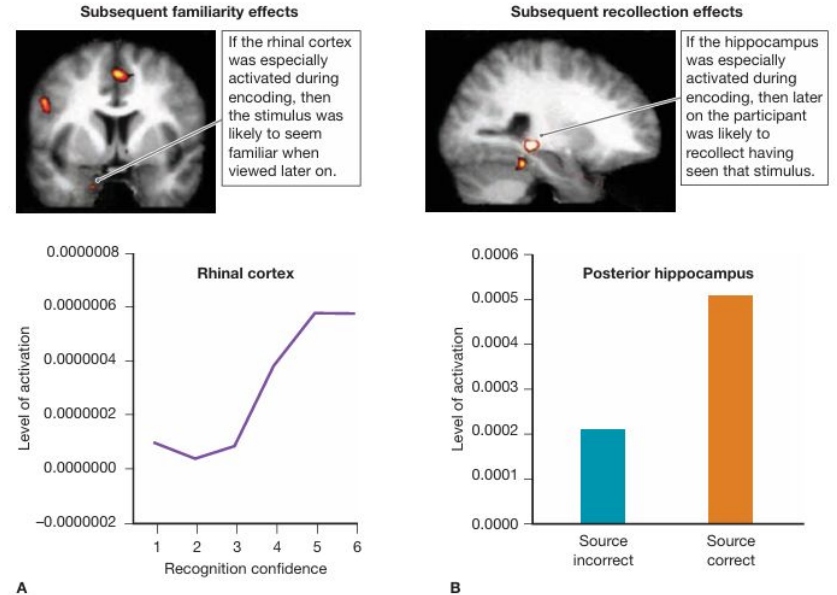
Types of Retrieval

- Recall
 - Free Recall
 - Cued Recall
- Recognition
 - Remember/Know judgements
 - Know - Vague **Familiarity**
 - Remember - **Source memory** intact
 - 2 lists A & B - recall words and check list recall confidence & correctness

Direct Memory Testing - Explicit memories

Indirect Memory Testing - Implicit memories (priming effects)

FIGURE 7.7 FAMILIARITY VERSUS SOURCE MEMORY



In this study, researchers tracked participants' brain activity during encoding and then analyzed the data according to what happened later, when the time came for retrieval. (AFTER RANGANATH ET AL., 2003)

Implicit Memory can be **Misleading**

1. **False Fame**

- a. Encoding - Participants asked to read aloud names (fake mixed)
- b. Retrieval (Immediate & 24hrs later) - “How famous is this person?” - New list with truly famous, non-famous real names, fake names from encoding, new fake names
- c. Finding - Fake names - did not call them famous, in immediate recall; 24 hr recall - fake names were mistaken to be famous
- d. Why? - Source Memory faded, but Familiarity lended credibility

2. **Illusion of Truth**

- a. Participants heard statements - some true, some false; with warning of the same
- b. Later, they were shown these statements with new statements (true)
- c. Finding - Repeated statements (even though false) were judged as true than new true statements
- d. Why? - Familiarity, not because evidence increased, no source memory about truthness of sentences

3. **Source memory confusion** - Mugshot error - familiar faces judged as criminals; eyewitness testimony

Relevance - Propaganda, misinformation effect, fake news

DRM Procedure

BED

WAKE

SNORE

REST

SNOOZE

NAP

AWAKE

BLANKET

PEACE

TIRED

DOZE

YAWN

DREAM

SLUMBER

DROWSY

Recall **DRM**

Pillow?

How confident?

Bed?

Dream?

Yawn?

Snore?

Sleep?

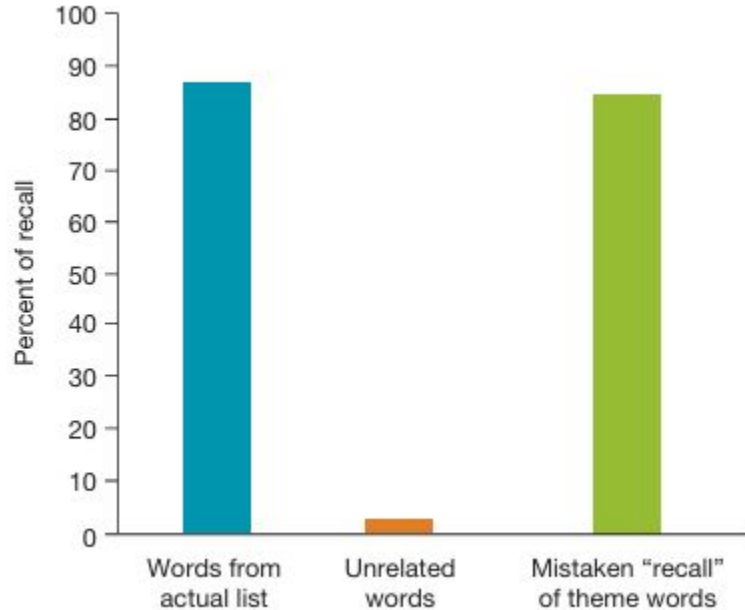


FIGURE 8.2 THE EFFECTS OF THE DRM PARADIGM

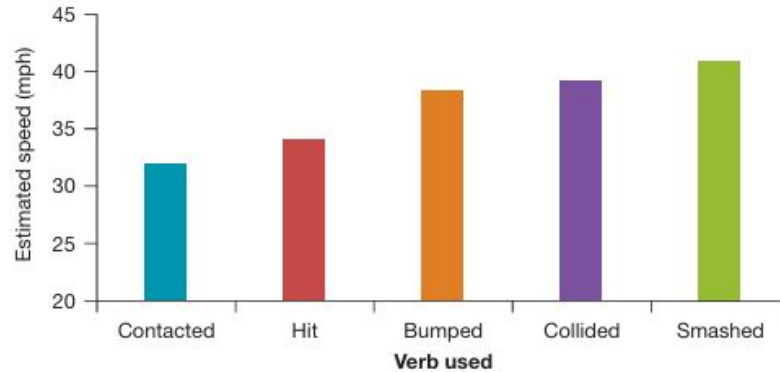
Because of the theme uniting the list, participants can remember almost 90% of the words they encountered. However, they're just as likely to "recall" the list's theme word — even though it was not presented.

Mechanisms leading to these errors are automatic thus people can't inhibit them

Schematic Knowledge

Planting False Memories

FIGURE 8.3 THE IMPACT OF LEADING QUESTIONS

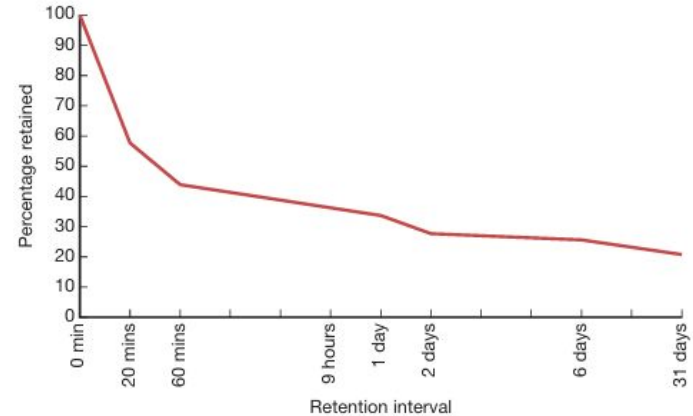


Witnesses who were asked how fast cars were going when they "hit" each other reported (on average) a speed of 34 miles per hour. Other witnesses, asked how fast the cars were going when they "smashed" into each other, gave estimates 20% higher. When all participants were later asked whether they'd seen broken glass at the scene, participants who'd been asked the "smashed" question were more likely to say yes—even though there was no broken glass. (AFTER LOFTUS & PALMER, 1974)

Forgetting

- **Decay theory of forgetting**
 - Memories fade with passage of time - connections not refreshed, thus weaken gradually
- **Interference theory**
 - New learning interferes with older learning
 - Passage of time not direct cause
 - Instead it creates opportunity for new learning, which disrupts old memories
 - **Proactive** - Old info interferes with new learning (you type old password instead of new one)
 - **Retroactive** - New info interferes with old memories (new perspective makes older outdated perspective inaccessible)
- **Retrieval Failure**
 - Forgotten memory still in LTM storage
 - But person unable to locate or retrieve it
 - E.g. likelihood your perspective has changed over increasing retention interval
 - TOT phenomenon - tip of the tongue

FIGURE 8.7 FORGETTING CURVE



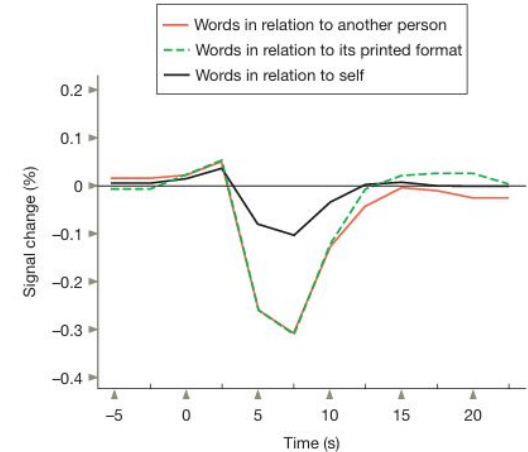
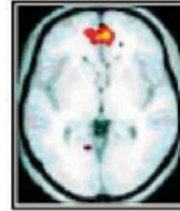
The figure shows retention after various intervals since learning. The data shown here are from classic work by Hermann Ebbinghaus, so the pattern is often referred to as an "Ebbinghaus forgetting curve." The actual speed of forgetting (i.e., how "steep" the "drop-off" is) depends on how well learned the material was at the start. Across most situations, though, the pattern is the same—with the forgetting being rapid at first but then slowing down. Mathematically, this pattern is best described by an equation framed in terms of "exponential decay."

Memory & Self

1. Overall info relevant to self is better remembered than non-self-relevant info - **“self-reference effect”**
2. But here too, memory errors
 - a. Your Memory is a mix of genuine recall & some schema-based reconstruction
 - b. Self-schema - there's an idea of consistency around the set of interwoven beliefs & memories that constitute a person's knowledge about themselves - **Bias**

FIGURE 8.9 SELF-REFERENCING AND THE BRAIN

MPFC activated during self-referential condition.



You are more likely to remember words that refer to *you*, in comparison to words in other categories. Here, participants were asked to judge adjectives in three conditions: answering questions like “Does this word describe the president?” or “Is this word printed in capital letters?” or “Does this word describe you?” Data from fMRI recordings showed a distinctive pattern of processing when the words were “self-referential.” Specifically, self-referential processing is associated with activity in the medial prefrontal cortex (MPFC). This extra processing is part of the reason why self-referential words are better remembered. (AFTER KELLEY ET AL., 2002)

Memory & Emotions

Memories are biologically **cemented in place**, by emotion's impact on memory consolidation

Emotional events trigger a response in amygdala & amygdala in turn increases activity in hippocampus

“Mull over emotional events” - i.e. memory rehearsal leads to better remembrance

Repressed & Recovered Memories - self protective mechanism (traumatic memories)

Flashbulb Memories



FLASHBULB MEMORIES

People often have especially clear and long-lasting memories for events like first hearing about Princess Diana's death in 1997, the attack on the World Trade Center in September 2001, or the news of Michael Jackson's death in 2009. These memories—called “flashbulb memories”—are vivid and compelling, but they are not always accurate.

What we do at the **MANDA Lab**

<https://www.mandalab.org>



Extended Reality studies of human memory

Temporal Context Models & Hippocampal subfields

Event Segmentation Theories

Traveling Waves

Memorability (Language & Memory)

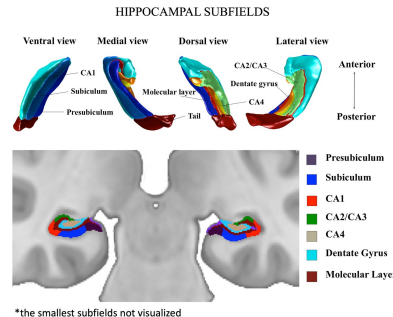
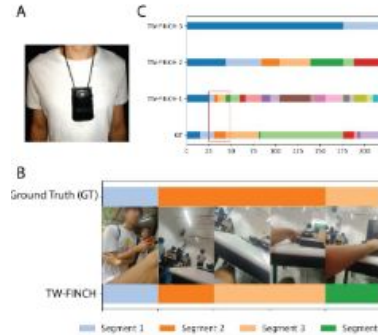
Neural dynamics & modeling

Neuromodulation

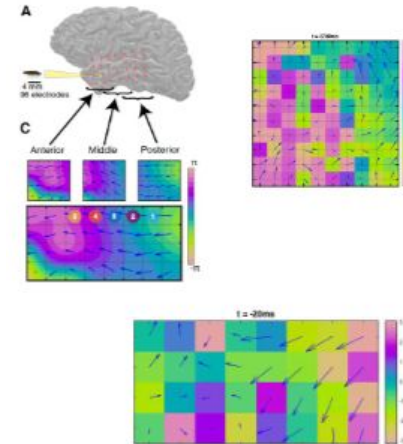
SDAM

Cognitive games

CPTSD



*the smallest subfields not visualized



Behavioral, FMRI, EEG, iEEG, TMS modalities

[Dr. Vishnu Sreekumar](#)